

astonishing breadth of knowledge. He was part of the class of mandarins that gained respectability in the late nineteenth century: upper middle class, powerful, but not exceedingly rich. His father was a doctor and professor of medicine, his uncle was a prominent scientist and administrator, and his cousin Raymond became a president of the republic of France. These were the days when the grandchildren of businessmen and wealthy landowners headed for the intellectual professions.

However, I can hardly imagine him on a T-shirt, or sticking out his tongue like in that famous picture of Einstein. There is something non-playful about him, a Third Republic style of dignity.

In his day, Poincaré was thought to be the king of mathematics and science, except of course by a few narrow-minded mathematicians like Charles Hermite who considered him too intuitive, too intellectual, or too “hand-waving.” When mathematicians say “hand-waving,” disparagingly, about someone’s work, it means that the person has: a) insight, b) realism, c) something to say, and it means that d) he is right because that’s what critics say when they can’t find anything more negative. A nod from Poincaré made or broke a career. Many claim that Poincaré figured out relativity before Einstein—and that Einstein got the idea from him—but that he did not make a big deal out of it. These claims are naturally made by the French, but there seems to be some validation from Einstein’s friend and biographer Abraham Pais. Poincaré was too aristocratic in both background and demeanor to complain about the ownership of a result.

Poincaré is central to this chapter because he lived in an age when we had made extremely rapid intellectual progress in the fields of prediction—think of celestial mechanics. The scientific revolution made us feel that we were in possession of tools that would allow us to grasp the future. Uncertainty was gone. The universe was like a clock and, by studying the movements of the pieces, we could project into the future. It was only a matter of writing down the right models and having the engineers do the calculations. The future was a mere extension of our technological certainties.

The Three Body Problem

Poincaré was the first known big-gun mathematician to understand and explain that there are fundamental limits to our equations. He introduced nonlinearities, small effects that can lead to severe consequences, an idea that later became popular, perhaps a bit too popular, as chaos theory. What’s so poisonous about this popularity? Because Poincaré’s entire point is about the limits that nonlinearities put on forecasting; they are not an invitation to use mathematical techniques to make extended forecasts. Mathematics can show